

Functions of many variables – 1

Partial derivatives: a survey of the subject

Homework

11 сентября 2026 года

Theoretical minimum. Differentiability means $f(x+h) = f(x) + \langle \nabla f(x), h \rangle + o(\|h\|)$ and is strictly stronger than the existence of all partial derivatives; continuity of the partial derivatives is what restores it. For a real-valued f the mean value theorem holds along a segment, for a map into $\mathbb{R}^m$ only the inequality $\|f(b) - f(a)\| \leq \sup \|Df\| \cdot \|b - a\|$. Mixed derivatives commute when they are continuous. Euler's identity $\langle x, \nabla f(x) \rangle = k f(x)$ characterises homogeneity of degree k , and comparing powers of t in $f(tx)$ splits a polynomial into homogeneous parts. A closed field is a gradient on a simply connected domain and need not be one otherwise. The Laplacian $\Delta = \partial_1^2 + \dots + \partial_n^2$ commutes with rotations and preserves the degree of a homogeneous polynomial. A relation valid for all values of the variables is to be differentiated in each of them separately, and a relation valid for all points of a family of figures is to be differentiated along a motion of that family.

1. (Putnam 2005) Let $P(x_1, \dots, x_n)$ denote a polynomial with real coefficients in the variables $x_1, \dots, x_n$ and suppose that $\Delta P(x_1, \dots, x_n) = 0$ (identically) and that $x_1^2 + \dots + x_n^2$ divides $P(x_1, \dots, x_n)$. Show that $P = 0$ identically.
2. (Putnam 2008) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function such that $f(x, y) + f(y, z) + f(z, x) = 0$ for all real numbers x, y and z . Prove that there exists a function $g: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x, y) = g(x) - g(y)$ for all real numbers x and y .
3. (Putnam 2009) Let f be a real-valued function on the plane such that for every square $ABCD$ in the plane $f(A) + f(B) + f(C) + f(D) = 0$. Does it follow that $f(P) = 0$ for all points P in the plane?

4. (Putnam 2009) Let $f: [0, 1]^2 \rightarrow \mathbb{R}$ be a continuous function on the closed unit square such that $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous on the interior $(0, 1)^2$. Let $a = \int_0^1 f(0, y)dy$, $b = \int_0^1 f(1, y)dy$, $c = \int_0^1 f(x, 0)dx$, $d = \int_0^1 f(x, 1)dx$. Prove or disprove: there must be a point (x_0, y_0) in $(0, 1)^2$ such that $\frac{\partial f}{\partial x}(x_0, y_0) = b - a$ and $\frac{\partial f}{\partial y}(x_0, y_0) = d - c$.
5. (Putnam 2010) Suppose that the function $h: \mathbb{R}^2 \rightarrow \mathbb{R}$ has continuous partial derivatives and satisfies the equation $h(x, y) = a \frac{\partial h}{\partial x}(x, y) + b \frac{\partial h}{\partial y}(x, y)$ for some constants a, b . Prove that if there is a constant M such that $|h(x, y)| \leq M$ for all $(x, y) \in \mathbb{R}^2$, then h is identically zero.
6. (Putnam 2011) Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable functions with the following properties:
- $F(u, u) = 0$ for every $u \in \mathbb{R}$;
 - for every $x \in \mathbb{R}$, $g(x) > 0$ and $x^2 g(x) \leq 1$;
 - for $(u, v) \in \mathbb{R}^2$, the vector $\nabla F(u, v)$ is either 0 or parallel to the vector $(g(u), -g(v))$.

Prove that there exists a constant C such that for every $n \geq 2$ and any $x_1, \dots, x_{n+1} \in \mathbb{R}$, we have $\min_{i \neq j} |F(x_i, x_j)| \leq \frac{C}{n}$.

7. (Putnam 2018) Let $f = (f_1, f_2)$ be a function from $\mathbb{R}^2$ to $\mathbb{R}^2$ with continuous partial derivatives $\frac{\partial f_i}{\partial x_j}$ that are positive everywhere. Suppose that

$$\frac{\partial f_1}{\partial x_1} \frac{\partial f_2}{\partial x_2} - \frac{1}{4} \left(\frac{\partial f_1}{\partial x_2} + \frac{\partial f_2}{\partial x_1} \right)^2 > 0$$

everywhere. Prove that f is one-to-one.

8. (Putnam 2019) Let $\mathcal{F}$ be the set of functions $f(x, y)$ that are twice continuously differentiable for $x \geq 1, y \geq 1$ and that satisfy the following two equations (where subscripts denote partial derivatives):

$$x f_x + y f_y = xy \ln(xy), \quad x^2 f_{xx} + y^2 f_{yy} = xy.$$

For each $f \in \mathcal{F}$, let $m(f) = \min_{s \geq 1} (f(s+1, s+1) - f(s+1, s) - f(s, s+1) + f(s, s))$. Determine $m(f)$ and show that it is independent of the choice of f .

9. (IMC 2002) Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function whose gradient $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}\right)$ exists at every point of $\mathbb{R}^n$ and satisfies the condition

$$\exists L > 0 \quad \forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\| \leq L\|x_1 - x_2\|.$$

Prove that $\forall x_1, x_2 \in \mathbb{R}^n \quad \|\nabla f(x_1) - \nabla f(x_2)\|^2 \leq L\langle \nabla f(x_1) - \nabla f(x_2), x_1 - x_2 \rangle$.

10. Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a differentiable function with continuous partial derivatives and with $f(0, 0) = 0$. Prove that there exist continuous functions $g_1, g_2: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $f(x, y) = xg_1(x, y) + yg_2(x, y)$.